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HYDROLOGICAL INVESTIGATIONS OF DINARIC KARST AT THE KRČIĆ CATCHMENT AND THE RIVER KRKA SPRINGS (YUGOSLAVIA)

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ABSTRACT

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A hydrological investigation of the relationship between the surface- and groundwater of the Krčić catchment and the spring water of the karst river, the Krka, is presented. The discharge along the Krčić decreases from the spring to the outlet (waterfall at the river Krka spring). Only in flood situations, some 30–60 days a year are normal hydrological relations found with discharge increasing towards the outlet. The Krčić river dries up regularly every year, with the number of days of zero flow increasing from the spring towards the outlet. A high degree of interdependence between groundwater levels in the Krčić catchment and the discharge of the Krka springs has been proved.

INTRODUCTION

Karst is characterized by its surface relief and a near-surface underground hydrographic network resulting from water circulation along soluble layers

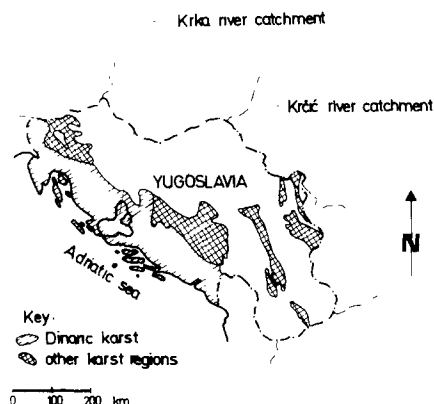


Fig. 1. Geographical situation of the Krka and Krčić catchment.

stations. The river Krka springs are located under the Krčić waterfall which is 40 m high. The subcutaneous zone is well-developed along the entire catchment and water flows rapidly underground through this zone. The average annual temperature of the region is ca. 12°C while the average yearly rainfall is 1100 mm, with a range from 800 to 1600 mm. The climate is continental, influenced by the mild Mediterranean climate. The summers are very hot, with air temperature as high as 35°C, whilst the winters are cold with temperatures as low as -5°C. The Krčić catchment is composed of Upper Triassic dolomites, Jurassic dolomites and limestone and Quaternary alluvial deposits. The Krčić flows partly through Upper Triassic dolomites and partly through the limestone overlying the dolomites.

THE DISCHARGE CONDITIONS ALONG THE KRČIĆ

The essential hydrological data for the gauging stations (Table 1) show how the strong karstification of the catchment affects the average annual discharges which decrease from the spring towards the outlet. Average daily

TABLE 1

Hydrological data along the Krčić river

gauging station	year of establishment	distance from the waterfall (m)	mean annual discharge [m ³ /s]					river
			1979	1980	1981	1982	1983	
1	1979	9910	76	8.1	5.5	5.6	3.6	KRČIĆ
2	1974	8230	7.2	7.7	5.8	5.0	3.1	-1-
3	1949	2550	6.0	6.2	4.9	4.1	3.0	-1-
4	1979	20	5.8	6.0	4.8	4.0	3.0	-1-
5	1897	500	14.5	17.0	16.1	15.5	16.7	KRKA

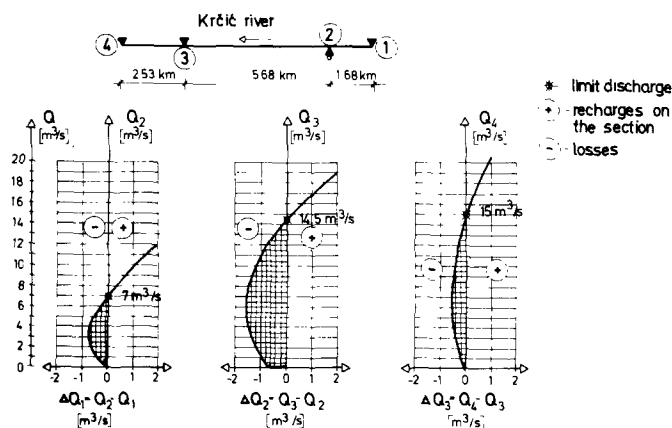


Fig. 3. Analysis of losses and recharges along the Krčić river (1979–1983).

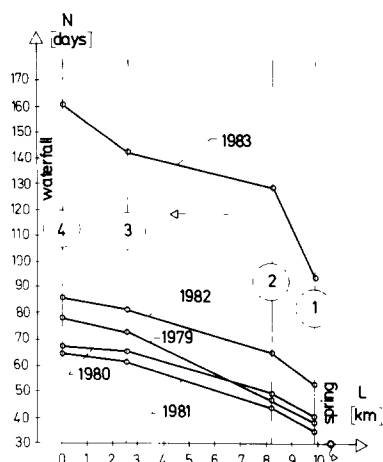


Fig. 4. Number of days with no flow N along the Krčić river.

discharges of the downstream station of the section i , for day j , ${}_jQ_{i+1}$, are plotted (Fig. 3) against the difference in discharge between the downstream and upstream sections ${}_j\Delta Q_i$, where:

$${}_j\Delta Q_i = {}_jQ_{i+1} - {}_jQ_i \quad (1)$$

When the value of ΔQ_i is negative, water is lost into the karst in section i ; the positive sign defines the hydrologically “normal” circulation. Further analyses indicate that during the short periods of “normal” circulation along the Krčić the zone of aeration of the open streamflow is completely filled by water. Figure 3 demonstrates the interactions of the curve of relation $Q_{i+1} - \Delta Q_i$ with the ordinate axis which do not significantly change from one year to another; these intersection points have been named “limit discharges”. The area defined by the curve and the ordinate axis is the negative domain (chequered in Fig. 3) and represents losses which are greatest on the second section, whereas the “limit discharges” along the Krčić range from 7 to $15 \text{ m}^3 \text{ s}^{-1}$. “Limit discharges” can be explained by the fact that losses due to the karstified catchment either decrease, completely disappear or become smaller than the lateral intercatchment inflow when the discharge is greater than the “limit discharge”. Thus, when the discharge of a particular section of the open flow exceeds the limit quantities, the Krčić catchment does not behave in a way typical of karst, and the discharge increases as the catchment area becomes larger. This can be explained by the assumption of saturation of the zone of aeration; such situations occur only after heavy rains and are of short duration. Simultaneous measurements carried out at numerous gauging stations confirm this assumption. Figure 4 gives the number of days on which the Krčić dried up in the period between 1979–1983. The drying begins downstream and gradually goes up to the spring. Thus even if the discharge at the spring is $1 \text{ m}^3 \text{ s}^{-1}$ there is no discharge at gauging station 3 and downstream from it.

Fourteen-year averages show that the Krčić river dries up for an individual station as follows: gauging station 1 (spring) — 46 days; station 2 — 57 days; station 3 — 80 days (min. 19—max. 143); station 4 (outlet; waterfall) — 90 days. The dry period begins most often at the beginning or middle of July and finishes by the middle or end of September; only twice in the winter period has the Krčić dried up. The beginning of the dry period is fairly precisely defined, while the end of the period ranges from September to February. The end of the dry period depends on the first heavy autumn or winter rains which exceed 80–100 mm within 24–72 h. The greatest discharge measured on the Krčić was $100 \text{ m}^3 \text{ s}^{-1}$, and it was recorded at station 3 which has the longest measurement record (since 1949).

THE RIVER KRKA SPRING

The Krka spring is a karst phenomenon which deserves special attention. Thoroughly studied from the geological and speleological viewpoint, the river Krka springs (Fig. 5) consist of three independent parts: (1) the Main Spring; (2) the Little Spring; and (3) the Third Spring. The Main Spring is located in the cave directly under the great waterfall and is thought to supply 80–90% of the total water quantities of all the Krka springs. The

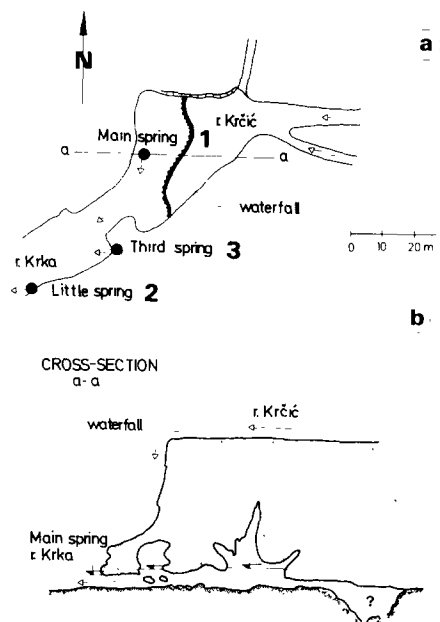


Fig. 5. The river Krka springs: (a) map; (b) cross-section of the speleologically investigated part of the cave with the Main Spring.

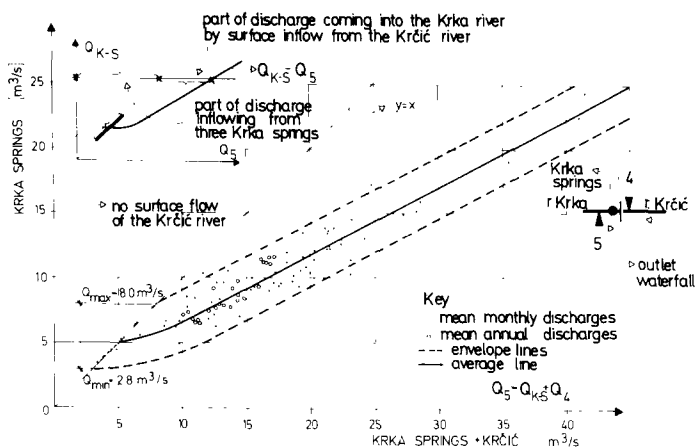


Fig. 6. Relationship between monthly mean and annual discharges of the Krka springs and the Krka station 5.

Third Spring has the smallest discharge and is situated on the left bank of the Krka 50 m downstream from the waterfall. The Little Spring, 150 m downstream of the waterfall, also on the left bank of the Krka river accounts for about 5–15% of all the underground discharge of the Krka river and its capacity is two to five times greater than the capacity of the Third Spring. Dye tracer tests of the groundwater from the upper horizons performed at eight upstream piezometers showed that these springs are hydrogeologically independent. The dye coming from particular piezometers appeared in only two springs; in no case was there any underground connection between the Little Spring and the upper horizons. The most interesting spring, considering its capacity, is the Main Spring located under the waterfall in a siphon (Fig. 5b). Speleologists have managed to gain access and take measurements. Thus the river Krka is formed partly by the surface flow of the Krčić coming across the waterfall and partly by the springs mentioned above. Its average annual discharge is $12.6 \text{ m}^3 \text{ s}^{-1}$. By subtracting the Krčić discharges measured at station 4 from the Krka discharges measured at station 5 the discharge of the Krka springs amounts to $7.5 \text{ m}^3 \text{ s}^{-1}$ over a several-years average. Figure 6 presents the relationship between the average monthly and yearly discharges of the Krka springs and total sum of discharges of the Krka springs and Krčić. The inset in the upper left corner provides an explanation of the plotted diagrams. The area between the ordinate axis (the Krka springs discharge) and the line $y = x$ represents the discharge inflowing into the Krka river exclusively from underground, whereas the second half of the quadrant represents the discharge supplied by surface flow from the Krčić. The points (plotted) on the line $y = x$ represent the situations when the discharge flows exclusively from underground, i.e. when there is no flow in the Krčić riverbed. Under these circumstances the discharge of the Krka river varies from 2.8 to $8 \text{ m}^3 \text{ s}^{-1}$.

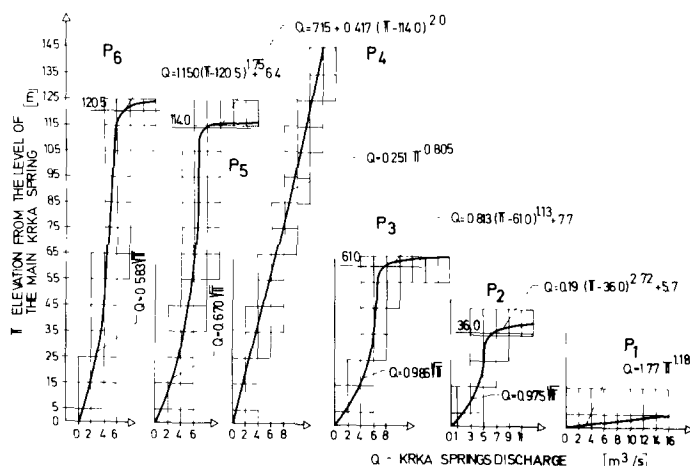


Fig. 7. Discharge curves of the Krka springs as a function of the groundwater level.

To define the discharge curves of the Krka springs as a function of the groundwater level of the Krčić catchment, observations were made of groundwater in the Krčić catchment. The general direction of the groundwater flow proved to be similar to that in the Krčić. Furthermore, some piezometers located on the dominant directions of the groundwater circulation react very quickly to the changes of water conditions in the rock, whilst changes of the water level in other piezometers occur more slowly and represent the normal average condition of the saturation of the karst. The groundwater level increases at the end of the summer period at an average of 20 m per day and at some piezometers amounts even to 40 m per day or 1.7 m h^{-1} ; this proves that the pore capacity does not exceed 2% of the total rock mass. In the Dinaric karst of Yugoslavia it ranges from 0.5 to 3% and in very rare cases, in small areas, exceeds 5%. A continuous water table was constructed from the recorded piezometer data. The maximum amplitude of groundwater oscillation at particular piezometers is about 100 m, with an average of 50 m. The groundwater level decreases at the beginning of the dry period at an average of 6 m per day, while the maximum observed velocity is 14 m per day or 0.6 m h^{-1} .

Analytical expressions for the discharge curves of the Krka spring have been defined following Hajdin and Ivetić (1978), Hajdin and Avdagić (1982) and Bonacci (1982). All these authors define the discharge curves from springs and swallow-holes in karst by applying essential hydraulic principles especially for turbulent flow with the quadratic law of resistance. These assumptions have proved applicable to the case of the Krka spring. Figure 7 presents the discharge curves of the Krka spring as a function of groundwater level as observed in the piezometers located in the Krčić catchment. Only six characteristic cases have been plotted although the computations were performed for all piezometers (28). At five piezometers presented in Fig. 7

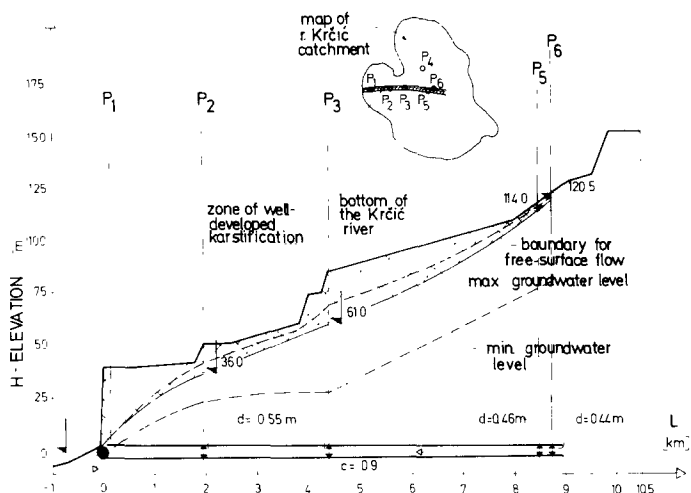


Fig. 8. Longitudinal cross-section of the Krčić riverbed with explanation of the conduit and diffuse flow in the Krčić catchment towards the Krka springs (d = dimension of hypothetical pipe).

(the exception being piezometer P_4) there is a height up to which the discharge has the same characteristics as the flow under pressure. Beyond this limit the discharge curve is formed according to the free-surface flow. In piezometer P_1 the discharge curve does not reach the limit and yet it has the parabolic shape characteristic of a flow with a free surface; as this piezometer is located near the Krka spring it does not actually measure flow under pressure. On the other hand, piezometer P_4 is very far away from the open flow of the Krčić (the same is true for the other piezometers far away from the Krčić) and the limiting level does not occur; these piezometers may well be situated outside the zone with free-surface flow. It may well be that quite close to the surface flow of the Krčić there is an underground flow which behaves as open flow. This flow could take place through pipes, though this is not likely, or through a system of intensely fractured rocks due to karstification. All equations for discharge curves were derived using five-day averages of groundwater levels and the respective discharges of the Krka springs because groundwater levels in the piezometers are observed only two or three times a week.

Figure 7 shows that the following general expression was used between the spring discharge Q , as a dependent variable, and the groundwater level π from the level of the Main Krka spring (in m), as an independent variable:

$$Q = A\sqrt{2g(\pi - H)} \quad (2)$$

where H is the spring level, and g is the acceleration of gravity. Parameter A can be defined as:

$$A = A^*c \quad (3)$$

where A^* represents the hydraulic measurement cross-section and c is the discharge coefficient always smaller than but close to 1; in karst it ranges from 0.8 to 0.95. Assuming that $c = 0.9$ and that the cross-section is a circular pipe (hardly possible in actual conditions) the diameter d of this pipe may be calculated as presented in Fig. 8. By analyzing the discharge curves of the Krka springs (Fig. 7) the underground water flow in the Krčić catchment towards the Krka springs may be characterized. The assumption of a pipe system and of the dimensions of the pipe is speculative: it is more likely to be a zone with fissures within which the flow can be approximated as flow under pressure. The water flow to the Krka spring is affected not only by infiltration but mostly by turbulent flow in karst "pipes" for which the quadratic law of resistance holds true.

CONCLUSION

The water losses along the Krčić caused by flow into the fractured karst medium through the zone of aeration have been quantified. Analytical expressions for the discharge curves as a function of the groundwater level in the karst hinterland have been defined. Two zones with turbulent conduit flow have been discovered; one, at the surface, can be partly classified as a well-developed subcutaneous zone, whereas the second is located about 30–100 m under the surface. Finally it should be emphasized that piezometers are of great importance in karst investigations. There should be as many as possible on each catchment; they provide data valuable for explaining the circulation of the groundwater in karst, and consequently for an optimal exploitation of the water potential.

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